

L5: Transformation of TRF Station Coordinates to CRF

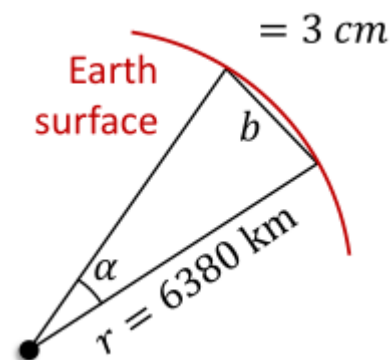
Variable Rotation:

- Earth's angular velocity not constant
- Orientation of reference axes varies
- Earth body changes position w.r.t. reference axis

Reason:

- Earth is oblate spheroid, not sphere
- Axes of rotation and figure axis do not coincide
- Earth neither rigid nor homogeneous

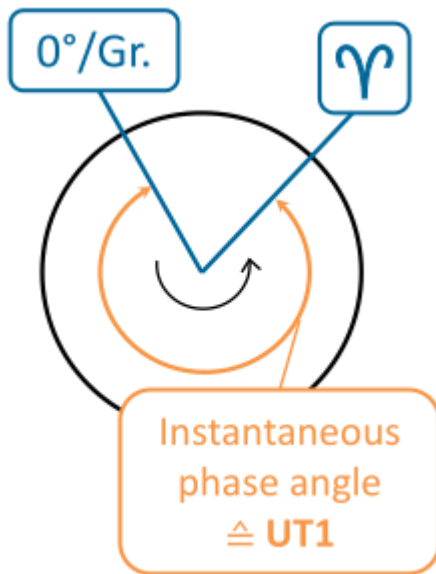
Meaning of MAS



$$b = \alpha[\text{rad}] \cdot r = 3\text{cm}$$

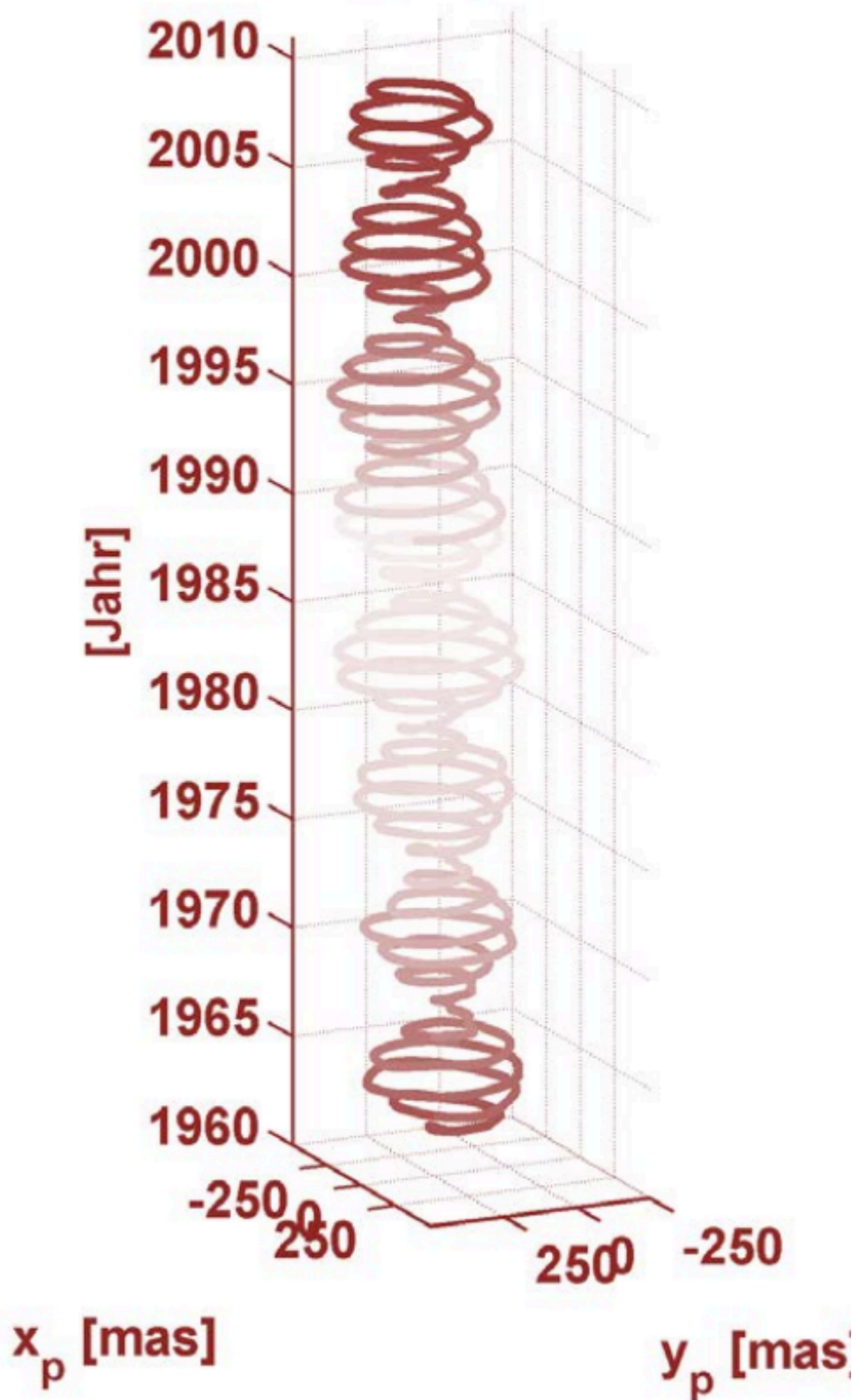
EOPs

- Orientation of the rotation axis with respect to CRF -> Precession/nutation
- Changing position of the rotation axis relative to the body-fixed RF -> Polar motion
- Variations in rotation rate (angular velocity) w.r.t. atomic frequency standard -> dUT1 (or ΔLOD)
- **5 Earth Orientation Parameters**



Polar Motion

Polbewegung 1962-2010,7



- Chandler Wobble
 - Free oscillation of Earth
 - Period ~433 days
- Annual Wobbles
 - Forced oscillation, main driver are annual pressure over continental areas

-> Beat

- Period of 6.3a and peak amplitudes of 9m

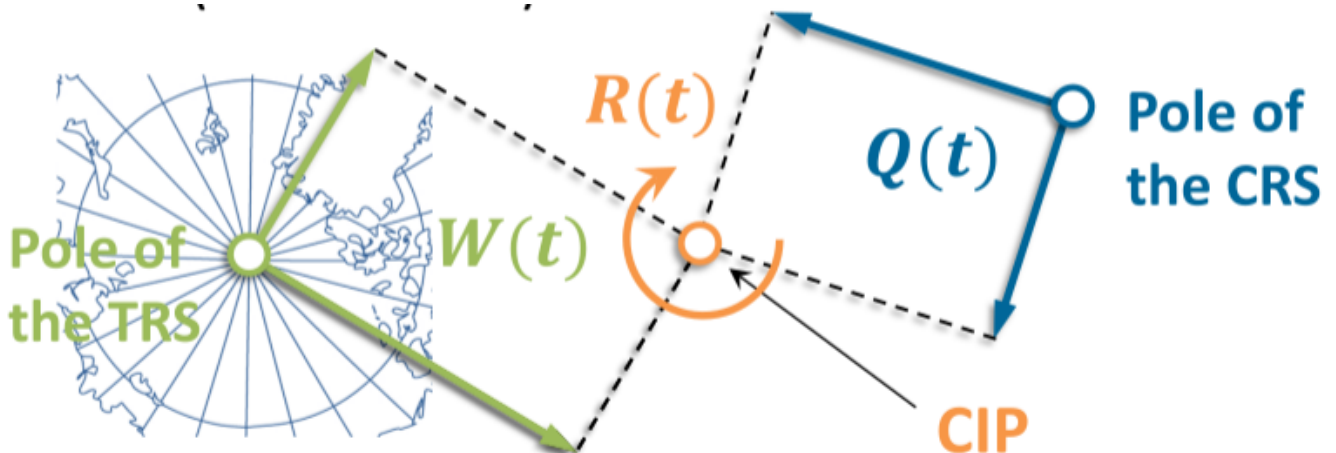
- Linear drift
 - 10cm/a toward 78°W
- Diurnal and semi-diurnal effects
 - ~3cm displacement of CIP at Earth's surface

UT- Δ LOD

- Variations of rotation rate expressed as deviations of UT1 from atomic time scale
 - $UT1 - UTC = \Delta UT1$
- Perturbations of nominal rotation rate $1/86400$ s
 - $\Delta LOD = -\frac{d}{dt} \Delta UT1$

Transformation TRS <-> CRS

- $[CRS] = Q(t) \cdot R(t) \cdot W(t) \cdot [TRS]$

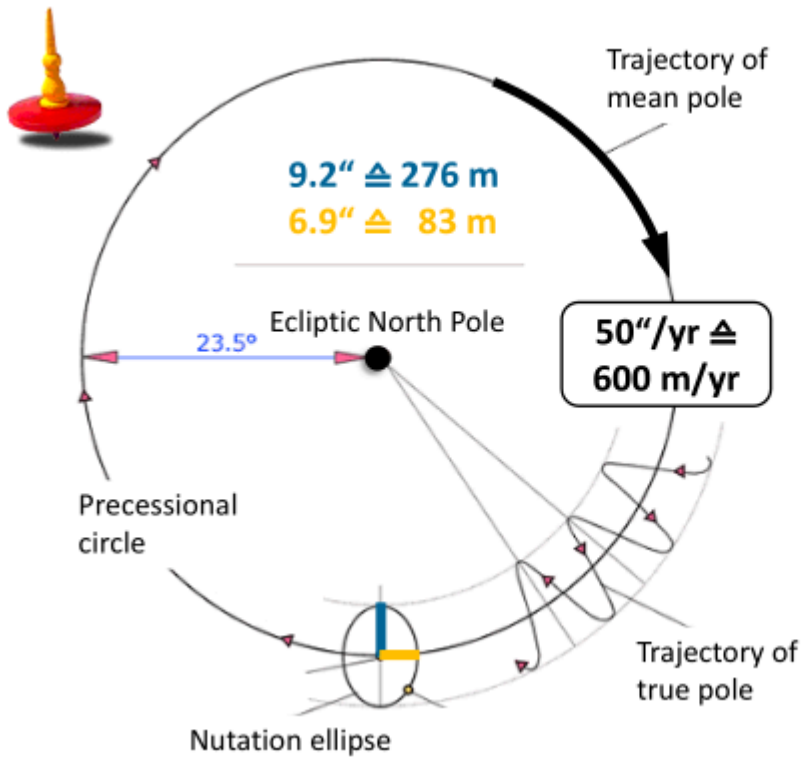


- At particular epoch
- Nutation $Q(t)$
 - $Q(t) = R_z(-E)R_y(-d)R_z(E + s)$
- Earth's Spin $R(t)$
 - $R(t) = R_z(-ERA)$
 - $ERA(T_u) = 2\pi(0.7790572732640 + 1.00273781191135448T_u)$
 - MJD of observation epoch 1.9.2020, 16:00UTC -> $T_u = mjd(UT1epoch) - 51544.5$
 - e.g. $mjd(UT1 epoch) = 59093 + 16/24 + dUT1/(24 \cdot 3600)$
- Polar motion $W(t)$

- $W(t) = R_z(-s')R_y(x_p)R_x(y_p)$

- -> ITRF
- With 5 Earth Orientation Parameters

Precession/Nutation



Precession period
also called "Great"
or "Platonic Year"

Nutation period
with largest
amplitude:
18.6 yr (Regression
of the lunar node)

